

AI Ecosystem for Fake Job Scam Detection and Fraud-Aware Job Recommendation

R26-SE-002

Project Proposal Report

Methmi N P C S – IT22188618

B.Sc. (Hons) Degree in Information Technology Specialized in
Software Engineering

Department of Software Engineering

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

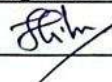
DECLARATION

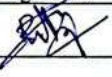
DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Methmi N PCS	IT22188618	മെത്തി

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Ms.Thilini Jayalath	14/03/2026	

Name	Date	Signature
Ms.Karthiga Rajendran	14/03/2026	

ABSTRACT (Maximum 300 words)

Online job platforms have become the primary medium for connecting job seekers with employment opportunities. Modern job recommendation systems commonly use techniques such as skill matching, collaborative filtering, and ranking optimization to improve recommendation relevance. However, many existing systems focus primarily on matching user profiles with job descriptions and do not consider the potential risks associated with fraudulent job postings. The increasing number of fake job advertisements, scam communications, and unreliable employers has created a serious trust and safety issue for job seekers. Therefore, ensuring the safety of recommended jobs has become an important challenge in job recommendation systems.

This research addresses the gap by proposing a risk-aware job recommendation and ranking model that integrates skill relevance with fraud risk analysis. The proposed system first performs skill matching between user profiles and job descriptions using TF-IDF and cosine similarity to identify candidate jobs. Then, fraud risk indicators obtained from multiple detection modules, including fake job detection, employer legitimacy analysis, and communication scam detection, are aggregated using the Entropy Weight Method to calculate a risk score. This risk score is converted into a safety score representing the trustworthiness of the job posting. Finally, the system applies the TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) multi criteria decision making method to rank jobs by considering both skill match scores and safety scores.

The expected outcome of this research is a job recommendation system that not only identifies relevant job opportunities but also prioritizes safer job postings, thereby reducing the risk of job scams. The proposed model can improve user trust in online job platforms and provide practical benefits for recruitment services. From a social perspective, it enhances the safety of job seekers, while commercially it offers potential integration into job portals and recruitment platforms to deliver trustworthy job recommendations.

Keywords: Job Recommendation System, Fraud Detection, Risk-Aware Ranking, TF-IDF, TOPSIS.

ACKNOWLEDGEMENT

I would like to sincerely thank my project supervisor Ms. Thilini Jayalath and co-supervisor Ms. Karthiga Rajendran and Lecturer in charge of the Resaerch Project Module Mr. Koliya Pulasinghe for their valuable guidance, support, and feedback throughout the preparation of this proposal. Their suggestions and advice helped me understand the research problem more clearly and improve the quality of this work.

I would also like to thank the lecturers of the department for providing the knowledge and learning environment that helped me develop the ideas for this research.

My appreciation also goes to my group members for their cooperation, discussions, and support during the development of this project.

Finally, I would like to thank my family and friends for their encouragement and support throughout this process. Their motivation helped me stay focused and complete this proposal successfully.

Table of Contents

DECLARATION	2
ABSTRACT (Maximum 300 words)	3
ACKNOWLEDGEMENT	4
LIST OF TABLES	7
LIST OF FIGURES	8
LIST OF ABBREVIATIONS	9
LIST OF APPENDICES	10
1. INTRODUCTION	11
1.1 Background and Context	11
1.2 Literature Review	12
1.3 Research Gap	13
2. OBJECTIVE	16
2.1 Main Objective	16
2.2 Specific Objectives	16
3. METHODOLOGY	17
3.1 Research Approach	17
3.2 Technical Methods and Algorithms	17
3.3 Tools and Platforms	20
3.4 Data Collection and Ethical Considerations:	20
4. HIGH-LEVEL SYSTEM ARCHITECTURE	21
4.1 Overview of the System	21
4.3 Data Flow Explanation	25
4.4 Scalability and Security Considerations	25
5. USER REQUIREMENTS	26
5.1 Stakeholder Identification	26
5.2 Functional Requirements	26
5.3 Non-Functional Requirements	27
6. COMMERCIALIZATION PLAN	27
7. BUDGET AND JUSTIFICATION	29
8. WORK BREAKDOWN STRUCTURE (WBS)	31
9. GANTT CHART	32
10. REFERENCES LIST	33

11. APPENDICES.....	34
----------------------------	-----------

LIST OF TABLES

Table 1: Comparison with existing projects.....	15
Table 2: Example Similarity Results.....	18
Table 3:Example conversion.....	18
Table 4:Example for Job Ranking and Recommendation.....	19
Table 5: Tools and Technologies	20
Table 6: Budget List.....	30

LIST OF FIGURES

Figure 1: Component Diagram	25
-----------------------------------	----

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CF	Collaborative Filtering
NLP	Natural Language Processing
TF-IDF	Term Frequency – Inverse Document Frequency
ML	Machine Learning
TOPSIS	Technique for Order Preference by Similarity to Ideal Solution
SDG	Sustainable Development Goal
DB	Database
UI	User Interface

LIST OF APPENDICES

Appendix 1: Data Flow Diagram.....	35
Appendix 2: Algorithm WorkFlow Diagram	36
Appendix 3: Login UI	37
Appendix 4: Sign UP UI.....	38
Appendix 5: Profile Setup UI	39
Appendix 6: Job Recommendation Dashboard	40
Appendix 7: Job Details Page	41
Appendix 8: Job Analysis Dashboard	Error! Bookmark not defined.
Appendix 9: Safe Job Ranking Results Page	42

1. INTRODUCTION

1.1 Background and Context

Today, many people use online job platforms to search for job opportunities. Websites such as job portals and recruitment platforms allow employers to post job vacancies and help job seekers easily find jobs that match their skills and interests. These platforms often use job recommendation systems to suggest suitable jobs to users based on their skills, experience, and preferences. This makes the job search process faster and more convenient for job seekers.

However, with the rapid growth of online job platforms, the number of fake job postings and job scams has also increased. Some scammers post fake job advertisements to collect personal information or money from job seekers. Many people, especially fresh graduates and individuals who are urgently looking for jobs, may apply to these job postings without realizing that they are fraudulent. This can lead to financial loss, data theft, and emotional stress for job seekers.

Most existing job recommendation systems mainly focus on matching user skills with job descriptions to recommend relevant jobs. These systems usually use techniques such as content-based filtering, collaborative filtering, or machine learning algorithms. Although these methods improve recommendation accuracy, they do not consider whether the job posting is safe or trustworthy. As a result, a job recommendation system may suggest a job that matches a user's skills but may actually be a fake or suspicious job posting.

To solve this problem, this research proposes a risk-aware job recommendation system that considers both job relevance and job safety. The system first finds jobs that match a user's skills using TF-IDF and cosine similarity techniques. After that, the system checks the risk level of each job using information from other fraud detection modules, such as fake job detection, employer legitimacy analysis, and communication scam detection. These risk indicators are combined to calculate a risk score, which is then converted into a safety score. Finally, the system ranks the jobs using the TOPSIS ranking method by considering both skill matching and safety score, and then recommends the safest and most relevant jobs to the user.

This research supports Sustainable Development Goal (SDG) 8: Decent Work and Economic Growth, which focuses on providing safe and productive employment opportunities. By helping users avoid job scams and identify safer job opportunities, this system can improve trust in online job platforms.

1.2 Literature Review

Many researchers have studied job recommendation systems to help job seekers find suitable job opportunities. These systems usually use machine learning techniques, recommender algorithms, and text similarity methods to match job seekers with relevant jobs based on their skills, experience, and preferences. However, most existing research mainly focuses on improving recommendation accuracy and does not consider the safety or fraud risk of job postings.

Bouneffouf (2013) proposed a Dynamic Risk-Aware Recommender System (DRARS) that considers the risk level of the user's situation when making recommendations [1]. The system uses a contextual bandit algorithm and adjusts the balance between exploration and exploitation based on the risk of the user's environment. The results showed that considering contextual risk can improve recommendation performance. However, this research focuses on general recommendation risk and does not address fraud risks related to job platforms.

Hybrid recommender systems are also commonly used in recommendation platforms. Çano and Morisio (2017) reviewed different hybrid recommender systems and explained that many systems combine collaborative filtering and content-based filtering techniques to improve recommendation accuracy [2]. These systems try to solve common problems such as data sparsity and cold start issues. However, most hybrid systems focus only on improving recommendation accuracy and do not consider security issues such as fraudulent job postings.

Collaborative filtering is another popular method used in job recommendation systems. Zhang et al. developed a job recommendation system based on user based and item based collaborative filtering algorithms [3]. Their system recommends jobs by analyzing similarities between users and job preferences. Experimental results showed that this method can improve recommendation accuracy. However, collaborative filtering methods often face problems such as the cold-start problem, and they do not check whether a job posting is legitimate or fraudulent.

Content-based recommendation techniques are also widely used in job platforms. Nadar et al. proposed a job recommendation system that uses TF-IDF and cosine similarity to compare job descriptions with user profiles. This system analyzes the text content of job descriptions and user skills to recommend suitable jobs [4]. Although this method improves job matching accuracy, it only focuses on relevance and does not verify the authenticity or safety of job postings.

Some research has also focused on resume analysis and automated skill extraction. The Skill Finder system automatically extracts skills from resumes and compares them with job requirements to find suitable job opportunities [5]. The system ranks candidates based on how well their skills match the job requirements. However, the system mainly focuses on skill matching and does not check whether the job postings or employers are trustworthy.

Similarly, resume analysis systems have been developed using machine learning techniques such as TF-IDF, cosine similarity, and K-Nearest Neighbors (KNN) [6]. These systems extract information from resumes and recommend suitable jobs based on similarity scores. Although these systems improve the recruitment process, they do not include methods to detect fraudulent job postings or scam communications.

Some studies have also focused on improving the ranking of recommended items. Zhang et al. proposed recommender models using matrix factorization techniques to improve ranking performance in recommender systems [7]. These methods improve the ranking accuracy of recommended items. However, they mainly focus on ranking quality and do not consider fraud risks when recommending jobs.

Trust aware recommender systems have also been studied to improve recommendation quality. Masrom et al. introduced a Trust-Aware Recommender System that uses trust relationships between users to improve recommendation accuracy [8]. However, this system mainly focuses on trust between users and does not analyze risks related to fraudulent job postings or employers.

Another study by Wegmeth et al. explored the selection of recommendation algorithms for improving ranking performance in recommender systems [9]. Their research showed that choosing the correct algorithm can significantly improve recommendation results. However, the study focuses mainly on algorithm performance and does not consider fraud risks in job recommendation systems.

Another approach used in decision-making and recommendation systems is multi-criteria ranking. Madanchian and Taherdoost introduced the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), a multi-criteria decision-making method used to rank alternatives based on their distance from an ideal solution and a negative ideal solution [10]. The TOPSIS method evaluates different alternatives by considering multiple criteria simultaneously and selecting the option that is closest to the optimal solution. This approach has been widely applied in various domains such as engineering, management, and information systems for decision support. However, most applications of TOPSIS focus on general decision-making problems and are not specifically designed to integrate fraud risk indicators into job recommendation systems.

Overall, most existing job recommendation systems mainly focus on matching user skills with job descriptions and improving recommendation accuracy. Very few studies consider the fraud risks related to job postings, employer legitimacy, or scam communications. Because of the increasing number of online job scams, it is important to develop a job recommendation system that considers both job relevance and job safety.

To address this problem, this research proposes a fraud-aware job recommendation and ranking model. The proposed system combines skill matching techniques with fraud risk indicators from multiple detection modules. By integrating skill relevance and fraud risk, the system aims to generate safer and more reliable job recommendations for users.

1.3 Research Gap

Many existing studies have focused on improving job recommendation systems by using techniques such as collaborative filtering, content-based filtering, and machine learning models. These systems mainly aim to recommend jobs that match the skills, experience, and preferences of job seekers. Although these approaches help improve recommendation accuracy, they mainly focus on relevance and do not consider whether the recommended jobs are safe or trustworthy.

Several research studies have also explored risk-aware recommendation systems, but most of them are developed for other domains such as finance or e-commerce. These systems are not specifically designed for job platforms and therefore do not consider the unique risks related to online recruitment, such as fake job postings, fraudulent employers, or scam communications.

Some studies have attempted to integrate risk into job recommendation systems, but they often consider only a limited number of risk factors. In many cases, these systems rely on a single risk indicator and do not combine multiple sources of fraud detection information. As a result, the overall risk assessment of job postings may not be accurate or reliable.

In addition, many existing job recommendation systems use ranking techniques that focus only on improving the accuracy of recommendations, without considering the safety of the recommended jobs. This means that a job with a high skill match score may still be recommended to users even if the job posting is suspicious or potentially fraudulent.

Therefore, there is a clear research gap in developing a job recommendation system that integrates both job relevance and fraud risk analysis. Current systems do not effectively combine skill matching, fraud detection indicators, and ranking algorithms within a single framework.

To address this limitation, this research proposes a risk aware ranking model for job recommendation systems that integrates skill matching with multiple fraud risk indicators, including fake job detection, employer legitimacy analysis, and communication scam detection. The system aggregates these risk indicators to calculate an overall risk score and then ranks jobs by considering both skill relevance and safety. This approach introduces a novel framework for safer job recommendations, helping job seekers avoid fraudulent opportunities while still receiving relevant job suggestions.

Functions / Features	Hybrid Recommender System (CF + Content-Based)	Collaborative Filtering Job Recommender	Content-Based Job Recommender (TF-IDF)	Risk-Aware Recommendation Model	Proposed Fraud-Aware Job Recommendation System
Skill Matching using Job Description	✓	✓	✓	✓	✓
Personalized Job Recommendation	✓	✓	✓	✓	✓
Fraud Risk Detection	✗	✗	✗	Limited	✓

Functions / Features	Hybrid Recommender System (CF + Content-Based)	Collaborative Filtering Job Recommender	Content-Based Job Recommender (TF-IDF)	Risk-Aware Recommendation Model	Proposed Fraud-Aware Job Recommendation System
Fake Job Post Detection	X	X	X	X	✓
Employer Legitimacy Analysis	X	X	X	X	✓
Communication Scam Detection	X	X	X	X	✓
Risk Score Calculation	X	X	X	Limited	✓
Safety Score for Job Recommendation	X	X	X	X	✓
Multi-Criteria Ranking Method	X	Limited	X	Limited	✓
Safe Job Recommendation	X	X	X	Partial	✓

Table 1: Comparison with existing projects

2. OBJECTIVE

2.1 Main Objective

To develop a risk-aware job recommendation and ranking model that integrates skill matching and fraud risk analysis in order to recommend relevant and safer job opportunities for job seekers.

2.2 Specific Objectives

- To perform skill matching between user profiles and job descriptions using TF-IDF and cosine similarity to identify relevant job opportunities.
- To collect and convert fraud detection results into risk indicators from multiple sources such as fake job detection, employer legitimacy detection, and communication scam detection.
- To calculate an overall risk score for each job posting using the Entropy Weight Method to represent the level of fraud risk.
- To convert the calculated risk score into a safety score to represent the trustworthiness of job postings.
- To rank candidate jobs using the TOPSIS ranking method by considering both skill match scores and safety scores.
- To generate final job recommendations that balance job relevance and safety for job seekers.

3. METHODOLOGY

3.1 Research Approach

This research follows a Design Science research approach because the main goal is to design and develop a new system that improves existing job recommendation systems. The study proposes a fraud-aware job recommendation and ranking model that integrates job relevance and fraud risk analysis.

The system is designed to recommend jobs not only based on the user's skills and preferences but also by considering the safety of job postings. The research focuses on developing a model that combines skill matching, fraud risk aggregation, and ranking algorithms to generate safer job recommendations.

The methodology consists of three main components:

- Skill Matching
- Risk Score Calculation
- Job Ranking and Recommendation

3.2 Technical Methods and Algorithms

Skill Matching:

The first step of the system is to identify jobs that match the user's skills.

The system collects user profile information, such as:

- skills
- experience
- job preferences
- education

Then the system retrieves job descriptions from the job database. Each job description includes information such as job title, required skills, responsibilities, and experience requirements.

Next, the text data is prepared using text preprocessing techniques, which include:

- converting text to lowercase
- removing punctuation
- removing stop words

After preprocessing, the system converts the text data into numerical values using the TF-IDF (Term Frequency Inverse Document Frequency) method.

TF-IDF helps identify important skill related words such as Python, SQL, and Machine Learning, while giving less importance to common words such as the, and, is.

Once the text is converted into vectors, the similarity between the user profile and each job description is calculated using Cosine Similarity.

Example similarity results:

Job	Similarity Score
Job A	0.90
Job B	0.85
Job C	0.80

Table 2: Example Similarity Results

These similarity scores are converted into skill match scores, which represent how well a job matches the user's skills.

Risk Score Calculation:

After identifying candidate jobs, the system evaluates the fraud risk associated with each job posting.

The system receives detection results from three fraud detection modules:

- Fake Job Post Detection Model
- Employer Legitimacy Detection Model
- Communication Scam Detection Model

These outputs are converted into numerical risk indicators.

Example conversion:

Detection Result	Risk Value
Legitimate / Not Scam	0
Suspicious	0.5
Fake / Scam	1

Table 3: Example conversion

The system then calculates the overall risk score using the Entropy Weight Method.

Risk Score Formula:

$$\text{Risk Score} = (w_1 \times \text{Fake Job Risk}) + (w_2 \times \text{Employer Risk}) + (w_3 \times \text{Communication Risk})$$

The entropy weight method automatically determines the importance of each risk indicator based on the available data.

The final risk score is normalized between **0 and 1**:

- 0: No risk detected
- Close to 1: High fraud risk

To make the score easier to use in the recommendation process, the risk score is converted into a safety score.

$$\text{Safety Score} = 1 - \text{Risk Score}$$

A higher safety score indicates a safer job posting.

Job Ranking and Recommendation:

After calculating the skill match score and safety score, the system ranks the candidate jobs.

Since the two scores have different ranges (skill score: 0–100, safety score: 0–1), the values are first normalized.

Next, the system creates a decision matrix using the normalized scores.

Example:

Job	Skill Score	Safety Score
Job A	0.90	0.17
Job B	0.85	1.00
Job C	0.80	0.67

Table 4: Example for Job Ranking and Recommendation

The ranking is performed using the **TOPSIS (Technique for Order Preference by Similarity to Ideal Solution)** algorithm.

TOPSIS identifies jobs that are closest to the ideal solution, which represents the best combination of high skill match and high safety score.

Finally, the system calculates a ranking score for each job and sorts them in descending order to generate the final job recommendations.

Example ranking result:

1. Job B
2. Job C
3. Job A

Even though Job A had the highest skill match score, it was ranked lower because its safety score was very low.

3.3 Tools and Platforms

The system will be developed using the following tools and technologies:

Tool	Purpose
Python	Implementation of algorithms and data processing
Scikit-learn	TF-IDF and cosine similarity calculations
Pandas / NumPy	Data processing and analysis
NLP libraries	Text preprocessing
Database (MySQL / MongoDB)	Storing user profiles and job data

Table 5: Tools and Technologies

The proposed system will be developed using Python and several supporting technologies because they provide strong support for machine learning, data processing, and text analysis. Python is selected as the main programming language due to its simplicity, flexibility, and wide range of libraries for data science and artificial intelligence applications. The Scikit-learn library will be used to implement techniques such as TF-IDF and cosine similarity, which are required for calculating skill similarity between user profiles and job descriptions. Pandas and NumPy will be used for data manipulation, numerical computation, and efficient handling of datasets during the preprocessing and analysis stages. Natural Language Processing (NLP) libraries will assist in text preprocessing tasks such as removing stop words, cleaning text, and preparing job descriptions and user profiles for analysis. In addition, a database system such as MySQL or MongoDB will be used to store user profiles, job postings, and system outputs in an organized way. These tools are widely used in machine learning and data science applications, making them suitable for developing the proposed fraud-aware job recommendation system.

3.4 Data Collection and Ethical Considerations:

Data Collection:

For this research, the system needs job posting data and user profile data to test the job recommendation model.

The job data will be collected from publicly available datasets, such as job posting datasets available on platforms like Kaggle. These datasets usually contain information such as job titles, job descriptions, required skills, company details, and experience requirements.

The system also needs user profile information to match jobs with users. Since real user data cannot be used due to privacy reasons, synthetic user profiles will be created for testing. These profiles will include details such as skills, education, experience, and job preferences to simulate real job seekers.

Before using the data, it will go through a data cleaning and preparation process. This includes removing duplicate records, handling missing values, converting text to lowercase, and removing unnecessary words. This step helps prepare the data so that it can be used properly by the recommendation algorithms.

Ethical Considerations:

Ethical issues are important when using data for research. In this study, only public datasets and synthetic data will be used. No real personal information will be collected from users.

All data will be used only for research and educational purposes. The system will also ensure that no personal identity information is stored or shared.

By following these steps, the research ensures that the data is used responsibly and ethically while developing the job recommendation system.

Validation Strategy and Performance Metrics:

The proposed system will be evaluated to measure its effectiveness in generating safe and relevant job recommendations.

Evaluation metrics may include:

- **Recommendation Accuracy** - How well the system matches jobs with user skills
- **Precision and Recall** - Quality of recommended jobs
- **Ranking Performance** - Effectiveness of the ranking algorithm
- **Safety Evaluation** - Ability to reduce risky job recommendations

The results of the proposed model will be compared with traditional job recommendation systems to demonstrate the benefits of integrating fraud risk analysis with recommendation algorithms.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed system is a Fraud Aware Job Recommendation System that recommends jobs based on both skill relevance and job safety. Most existing job recommendation systems only match user skills with job descriptions. However, this system also checks whether a job posting is safe or risky before recommending it to users.

The system works through several components that process user information, analyze job postings, evaluate fraud risk, and finally rank the jobs.

4.1 Overview of the System

The system mainly includes the following components:

1. User Profile and Job Descriptions
2. Skill Matching Module
3. Candidate Job Selection
4. Fraud Risk Indicator Module
5. Risk Aggregation Module
6. Safety Score Calculation
7. Ranking Engine
8. Safe Job Recommendations

Each component performs a specific task to generate relevant and safe job recommendations.

1. User Profile and Job Descriptions

The system first collects two types of information.

User profile information

- Skills
- Work experience
- Education
- Job preferences

Job description information

- Job title
- Required skills
- Responsibilities
- Experience requirements

This information is used as the input for the recommendation process.

2. Skill Matching Module

The skill matching module compares the user profile with job descriptions to find jobs that match the user's skills.

This module uses two techniques:

- **TF-IDF**
- **Cosine Similarity**

TF-IDF converts the text data (such as resumes and job descriptions) into numerical values. Cosine similarity then measures how similar the user profile is to each job description.

The output of this step is a skill match score that shows how well a job matches the user's skills.

3. Candidate Job Selection

After calculating the similarity scores, the system selects the top matching jobs as candidate jobs.

These jobs are relevant to the user's skills, but they may still be unsafe or fraudulent. Therefore, the system needs to evaluate the risk level of these jobs.

4. Fraud Risk Indicators

In this step, the system receives risk information from three fraud detection modules:

- Fake Job Detection Module
- Employer Legitimacy Detection Module
- Communication Scam Detection Module

Each module analyzes a different part of the job posting and gives a result such as:

- Legitimate
- Suspicious
- Fake or Scam

These results are converted into risk indicator values.

5. Risk Aggregation Module

The system combines all risk indicators using the Entropy Weight Method.

This method calculates the importance (weight) of each risk indicator and produces an overall risk score for each job.

The risk score ranges between 0 and 1.

- 0 → No fraud risk
- 1 → High fraud risk

6. Safety Score Calculation

To make the risk information easier to use, the system converts the risk score into a safety score.

Safety Score = 1 – Risk Score

If the safety score is close to **1**, the job is considered safe.

If the safety score is close to **0**, the job may be risky.

7. Ranking Engine

The ranking engine is the main component of this research.

It ranks jobs based on two factors:

- Skill Match Score
- Safety Score

The ranking is performed using the TOPSIS algorithm.

TOPSIS finds jobs that are closest to the ideal job, which means the job with the highest skill match and highest safety score.

Each job receives a ranking score, and the system sorts them from highest to lowest.

8. Safe Job Recommendations

Finally, the system shows the final list of recommended jobs to the user.

These recommended jobs:

- Match the user's skills
- Have lower fraud risk
- Are ranked using the ranking algorithm

This helps users find relevant and safer job opportunities.

4.2 Component Interaction

The proposed system consists of several components that work together to detect job scams and recommend safe jobs to users. My component, which is the Fraud-Aware Job Recommendation and Ranking Model, interacts with other components in the system to combine skill matching and fraud risk information.

First, the system receives user profile information and job descriptions as inputs. The user profile includes details such as skills, education, experience, and job preferences. The job descriptions contain information like job title, required skills, responsibilities, and company details. My component uses this information to perform skill matching between the user profile and job descriptions using TF-IDF and cosine similarity. This step calculates a skill match score and selects the most relevant jobs as candidate jobs.

After the candidate jobs are selected, my component receives risk indicators from the other modules in the system. These include the Fake Job Detection Model, the Employer Legitimacy Verification Model, and the Scam Communication Detection Model. Each of these modules analyzes a different part of the recruitment process and provides outputs such as job fraud risk, employer trust score, and communication scam risk.

My component then combines these risk indicators to calculate an overall risk score using the Entropy Weight Method. This score represents how risky a job posting is. The system then converts the risk score into a safety score, which shows how safe the job is for the user.

Finally, my component uses the TOPSIS ranking algorithm to rank the candidate jobs. This ranking considers both the skill match score and the safety score. Jobs that have high skill relevance and low fraud risk will receive higher ranking scores.

The final result is a list of safe and relevant job recommendations for the user. Through this process, my component connects the job recommendation part of the system with the fraud

detection modules to ensure that users receive job recommendations that are both suitable and safe.

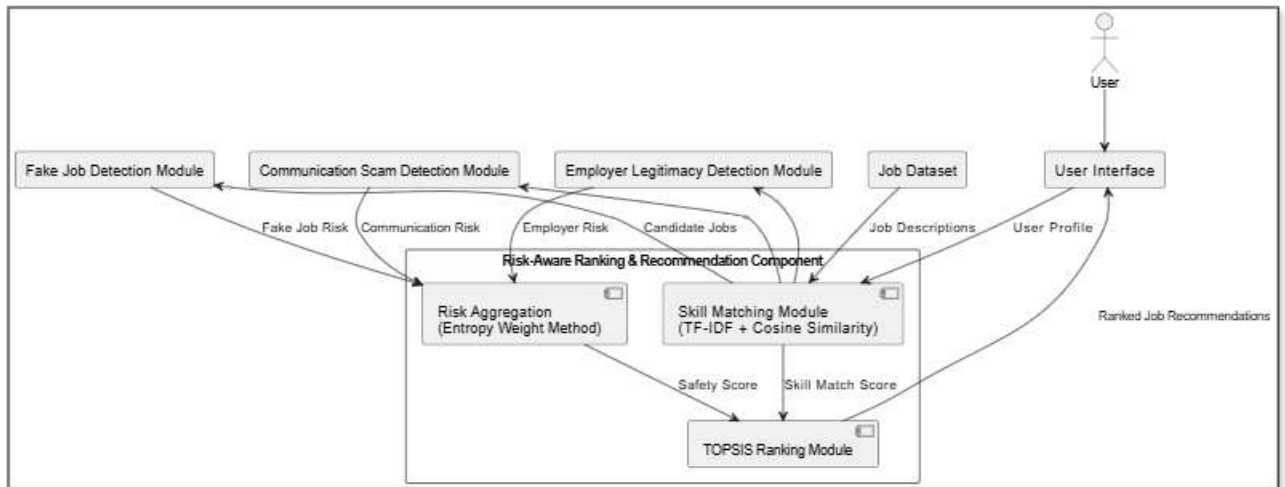


Figure 1: Component Diagram

4.3 Data Flow Explanation

The data flows step by step through the system. First, the system receives user information and job descriptions. Then it calculates skill similarity, evaluates fraud risks, calculates safety scores, and finally ranks the jobs to generate safe job recommendations.

4.4 Scalability and Security Considerations

The system is designed to handle large numbers of job postings and user profiles. Using efficient data processing tools allows the system to analyze large datasets.

To ensure security, the system will avoid storing sensitive personal information and will use anonymized or synthetic data for testing. This helps protect user privacy and ensures responsible data usage.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The main stakeholders of this system are the people or organizations that interact with or benefit from the job recommendation system.

1. Job Seekers

Job seekers are the main users of the system. They use the platform to find job opportunities that match their skills and preferences. They also benefit from receiving safer job recommendations that reduce the risk of job scams.

2. System Administrators

System administrators manage the system and maintain the job database. They ensure that the system runs smoothly and that job postings are updated properly.

3. Employers / Recruiters

Employers or recruiters can post job opportunities on the platform. The system helps connect them with suitable candidates whose skills match the job requirements.

4. Platform Developers / Researchers

Developers and researchers are responsible for designing, developing, and improving the recommendation system and fraud detection models.

5.2 Functional Requirements

The system should perform the following functions:

FR1: The system should allow users to create and manage their job seeker profiles.

FR2: The system should collect user information such as skills, experience, education, and job preferences.

FR3: The system should store and manage job descriptions in the job database.

FR4: The system should perform skill matching between user profiles and job descriptions using text similarity techniques.

FR5: The system should identify candidate jobs based on similarity scores.

FR6: The system should receive fraud risk indicators from fake job detection, employer legitimacy detection, and communication scam detection modules.

FR7: The system should calculate a risk score for each job posting using the entropy weight method.

FR8: The system should convert the risk score into a safety score.

FR9: The system should rank candidate jobs using the TOPSIS ranking algorithm.

FR10: The system should generate a ranked list of safe job recommendations for users.

5.3 Non-Functional Requirements

Security

- The system should protect user data and avoid storing sensitive personal information.
- User information should be handled securely to protect privacy.

Performance

- The system should process job recommendations efficiently even with a large number of job postings.
- The recommendation results should be generated within a reasonable time.

Usability

- The system interface should be simple and easy for job seekers to use.
- Users should be able to easily understand the recommended job results.

Scalability

- The system should be able to handle an increasing number of users and job postings.

Reliability

- The system should provide consistent and accurate job recommendations without errors.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona:

The main target market for this system is online job platforms, recruitment companies, and job seekers. Many job portals face problems with fake job postings and recruitment scams. This system can help these platforms provide safer job recommendations for their users.

The main users of this system include:

Job Seekers – People who are looking for jobs and want reliable job recommendations that match their skills and are safe.

Job Platforms / Recruitment Websites – Companies that provide job search services and want to improve the quality and safety of job recommendations.

Recruitment Agencies – Organizations that help employers find suitable employees and want to avoid fake job postings.

Value Proposition:

The main value of this system is that it provides job recommendations that are both relevant and safe.

Most job recommendation systems only match skills with job descriptions. They do not check whether a job posting is fake or risky. This system solves that problem by checking fraud risks before recommending jobs.

This helps:

- Reduce job scams
- Increase trust in job platforms
- Help job seekers find safer job opportunities

Revenue Model:

The system can generate income in several ways.

Subscription Model

Job platforms or recruitment companies can pay a monthly or yearly subscription to use the system.

API Integration

The system can be offered as an API that job platforms can connect to their existing systems.

Premium Features

Job seekers may pay for advanced recommendation features or detailed job safety analysis.

Enterprise Licensing

Large recruitment companies can buy licenses to use the system in their own platforms.

Cost Estimation:

The main costs for this system include:

- System development costs
- Cloud hosting and database storage
- Machine learning model development
- System maintenance and updates

There will be initial costs for building the system and ongoing costs for maintaining the servers and improving the models.

Pricing Strategy:

The system can use different pricing plans based on the features provided.

For example:

Basic Plan - Basic job recommendation features.

Professional Plan - Includes fraud detection and safer job ranking.

Enterprise Plan - Full system integration with advanced features for large companies.

The price can depend on the number of users or job postings processed by the system.

Competitive Advantage:

The main advantage of this system is that it focuses on both job relevance and job safety.

Most existing job recommendation systems only match skills with job descriptions. They do not consider the risk of fraudulent job postings.

This system combines:

- Skill matching using TF-IDF and cosine similarity
- Fraud risk analysis from multiple sources
- Risk score calculation using the entropy weight method
- Job ranking using the TOPSIS algorithm

Because of this, the system can recommend jobs that are both relevant and safer, which makes it different from traditional job recommendation systems.

Intellectual Property Considerations:

The proposed system includes a unique model that combines job recommendation and fraud risk analysis.

The system design, algorithms, and software code can be protected as intellectual property. This may include:

- Copyright protection for the software code
- Protection of the system architecture and algorithms
- Possible patent opportunities for the fraud-aware ranking model

Proper documentation and licensing will help protect the system if it is commercialized in the future.

7. BUDGET AND JUSTIFICATION

The budget mainly includes costs related to hardware, cloud services, and research activities. A computer is required to develop and test the recommendation algorithms. Cloud services are used to store datasets and perform data processing tasks. Most software tools used in the project are open-source, which reduces the overall cost. Human effort cost represents the time and work required to design and implement the system. Miscellaneous costs cover internet usage and other small research related expenses.

No	Item	Description	Estimated Cost (LKR)	Justification
1	Hardware	Laptop / Computer for system development and testing	0	Required for developing, testing, and running machine learning models and algorithms.
2	Software Tools	Development tools (Python, VS Code, libraries)	0	Most development tools used in this research are open-source and freely available.
3	Cloud Services	Cloud storage and computing services (AWS / Google Cloud)	-	Used to store datasets and run experiments for job recommendation models.
4	Dataset Preparation	Public job datasets and data preprocessing tools	7500	Required for collecting and preparing job posting datasets used in the experiments.
5	Human Effort	Time spent on research, development, and testing	75000	Represents the effort required to design, implement, and evaluate the proposed system.
6	Miscellaneous	Internet, documentation, and other research expenses	5000	Covers additional small costs related to research activities.

Table 6: Budget List

8. WORK BREAKDOWN STRUCTURE (WBS)



Figure 2: Work Breakdown Structure

9. GANTT CHART



Figure 3: Gantt Chart

10.REFERENCES LIST

- [1 D. Bouneffouf, "DRARS, A Dynamic Risk-Aware Recommender System," 2013. [Online].
] Available: <https://theses.hal.science/tel-01026136/document>.
- [2 E. Çano and M. Morisio, "HybridRecommenderSystems," 2017. [Online]. Available:
] https://www.researchgate.net/publication/321281871_Hybrid_Recommender_Systems_A_Systematic_Literature_Review.
- [3 C. Y. Z. N. Yingya Zhang, 2014. [Online]. Available:
] <https://ieeexplore.ieee.org/document/7064250>.
- [4 A. S. S. R. D. K. M. Thansidh Kirthan Nadar. [Online]. Available:
] https://www.researchgate.net/publication/385194961_Job_Recommendation_by_Content_Filtering_using_TF-IDF_and_Cosine_Similarity.
- [5 T. R. Kalva, 2013. [Online]. Available: <https://digitalcommons.usu.edu/gradreports/343/>.
]
- [6 K. N. J. L. G. T. R. M. A. H. S. S. K S VARSHITH REDDY, 2025. [Online]. Available:
] <https://www.irejournals.com/formatedpaper/1707439.pdf>.
- [7 H. B. S. Y. Richong ZHANG, 2016. [Online]. Available:
] https://www.researchgate.net/publication/282477414_Recommender_systems_based_on_ranking_performance_optimization.
- [8 N. K. A. A. R. A. A. R. S. Masrom, 2018. [Online]. Available:
] https://www.researchgate.net/publication/330193192_An_Evaluation_Framework_of_Trust_Aware_Recommender_System.
- [9 T. V. J. B. Lukas Wegmeth, 2024. [Online]. Available:
] https://www.researchgate.net/publication/383911715_Recommender_Systems_Algorithm_Selection_for_Ranking_Prediction_on_Implicit_Feedback_Datasets.
- [1 H. T. Mitra Madanchian, 2020. [Online]. Available:
0] https://www.researchgate.net/publication/374540287_A_comprehensive_guide_to_the_TOPSIS_method_for_multi-criteria_decision_making.

11.APPENDICES

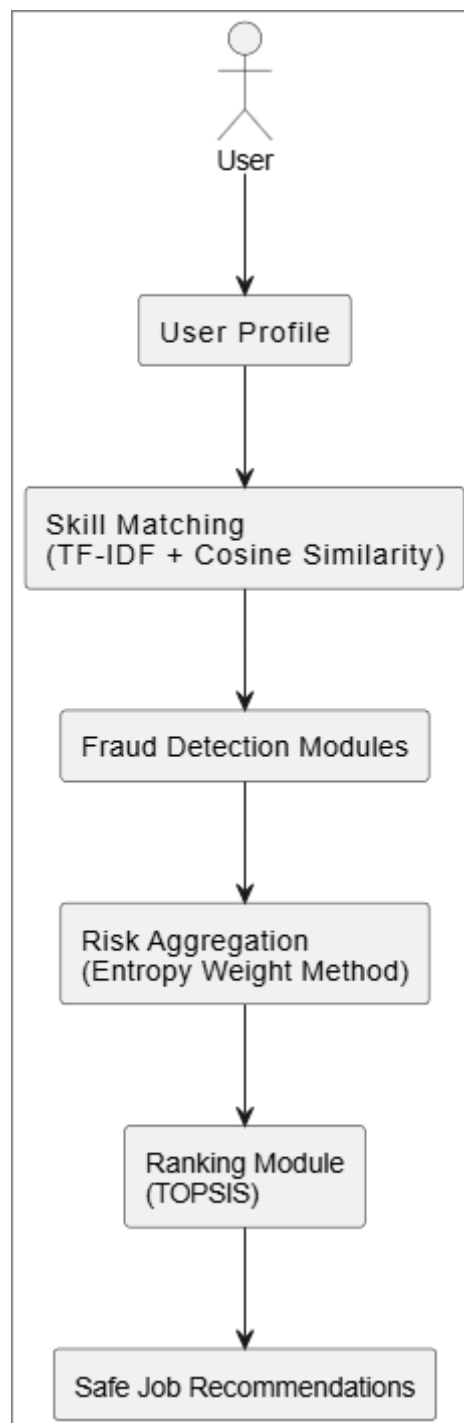
11.1 Survey instruments

The questionnaire used to gather user input on recruiting online experiences and employer verification requirements is included in this appendix. The purpose of the survey was to gather requirements for the planned employer-legitimacy verification system and to understand users' concerns about job scams.

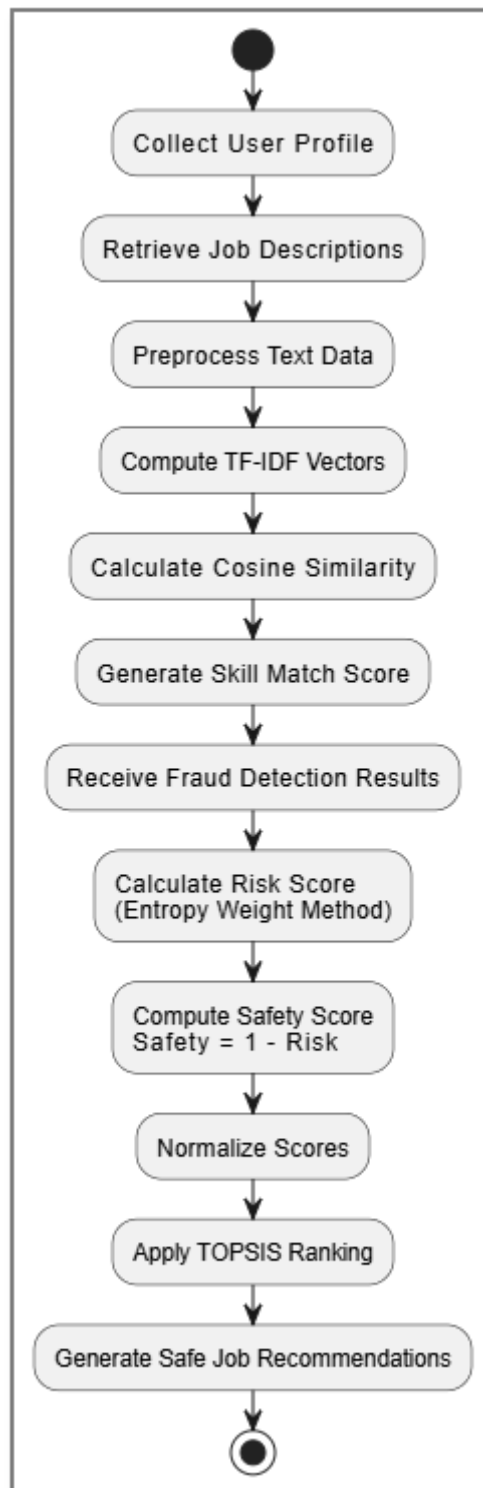
The questions included:

- What is your group age?
- What is your current status?
- Have you ever searched for jobs online?
- Which platforms do you used to search for jobs?
- Have your ever seen fake or suspicious job advertisements online?
- What characteristics make a job advertisement look suspicious?
- Do you usually check the company details applying for a job?
- Which company-related factors increase your trust in a job opportunity?
- Have you ever received suspicious messages from someone claiming to be a recruiter?
- Through which platforms did you receive suspicious recruiter messages?
- What type of suspicious behavior did you notice in those messages?
- Have you ever been asked to pay money for a job opportunity?
- How confident are you in identifying fake job offers or recruitment scams?
- Do you think recruitment scams are increasing on online platforms?
- Would you use a system that automatically detects scam job messages or suspicious recruiters?
- Which features would help you detect job scams more effectively?
- Do you think AI technology can help reduce recruitment scams and protect job seekers?
- In your opinion, what is the biggest risk when applying for jobs online?

11.2 Detailed diagrams

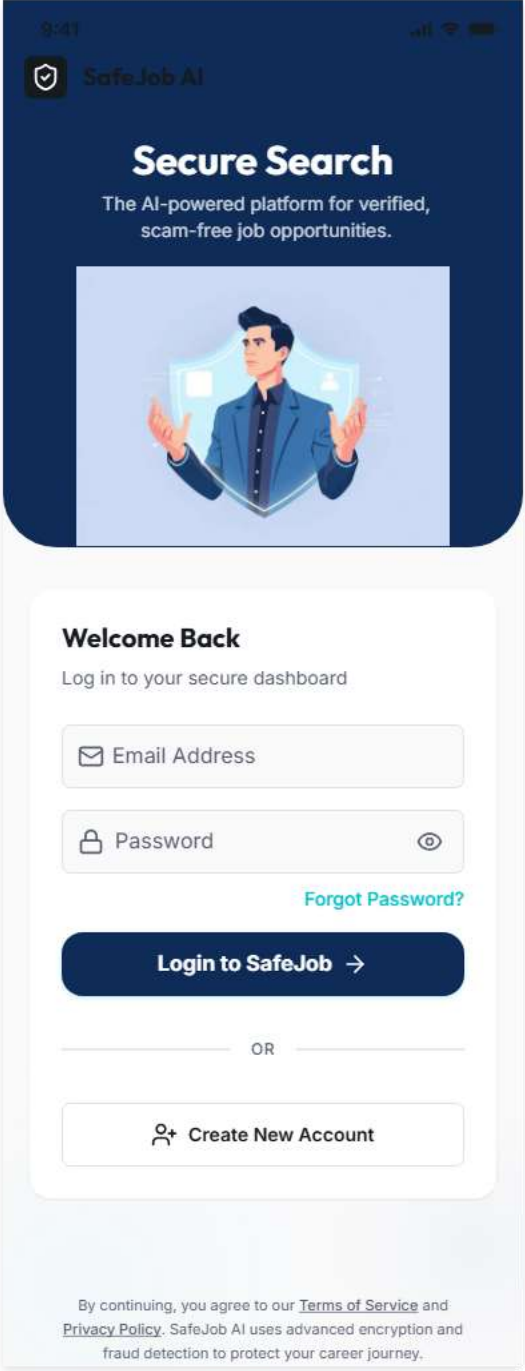


Appendix 1: Data Flow Diagram



Appendix 2: Algorithm WorkFlow Diagram

11.3 Preliminary UI mockups



The image shows a mobile app login screen for 'SafeJob AI'. At the top, the status bar shows the time 9:41 and signal strength. The app header features a shield icon and the text 'SafeJob AI'. Below this is a 'Secure Search' section with a subtitle 'The AI-powered platform for verified, scam-free job opportunities.' and an illustration of a man in a suit holding a glowing shield. The main login area is titled 'Welcome Back' and includes the instruction 'Log in to your secure dashboard'. It contains two input fields: 'Email Address' with an envelope icon and 'Password' with a lock icon and a toggle eye icon. A 'Forgot Password?' link is positioned below the password field. A dark blue 'Login to SafeJob' button with a right arrow is below the inputs. A horizontal line with 'OR' in the center separates the login section from the 'Create New Account' button, which features a person icon and a plus sign. At the bottom, a small disclaimer states: 'By continuing, you agree to our [Terms of Service](#) and [Privacy Policy](#). SafeJob AI uses advanced encryption and fraud detection to protect your career journey.'

9:41

SafeJob AI

Secure Search

The AI-powered platform for verified, scam-free job opportunities.

Welcome Back

Log in to your secure dashboard

Email Address

Password

[Forgot Password?](#)

Login to SafeJob →

OR

Create New Account

By continuing, you agree to our [Terms of Service](#) and [Privacy Policy](#). SafeJob AI uses advanced encryption and fraud detection to protect your career journey.

Appendix 3: Login UI

9:41

<

Create Account



Join the Network

Protect your career path with AI-driven fraud detection.



FULL NAME



 John Doe

EMAIL ADDRESS



 john@example.com

PASSWORD



STRENGTH

MEDIUM

CONFIRM PASSWORD



☐ I agree to the [Terms of Service](#) and acknowledge the [Privacy Policy](#) regarding fraud protection data.

Create Account →

Already have an account? [Log In](#)



 PROTECTED BY SAFEJOB AI ANTI-FRAUD ENGINE

By registering, you agree to our automated fraud analysis processes which help keep the job market secure for all participants.

Appendix 4: Sign UP UI

9:41

Match: 85%

Profile Setup

Complete your profile

This helps us verify your identity & match jobs.

Step 2/3

Skills & Expertise

Cybersecurity

Python

Threat Analysis

+ Add Skill

Q Search for skills (e.g. Java, SQL)...

Education

Master's Degree

Experience

3

Security Analyst

Preferred Location

Remote

New York, NY

San Francisco, CA

London, UK

Resume / CV

Tap to upload your resume

PDF, DOCX (Max 5MB)

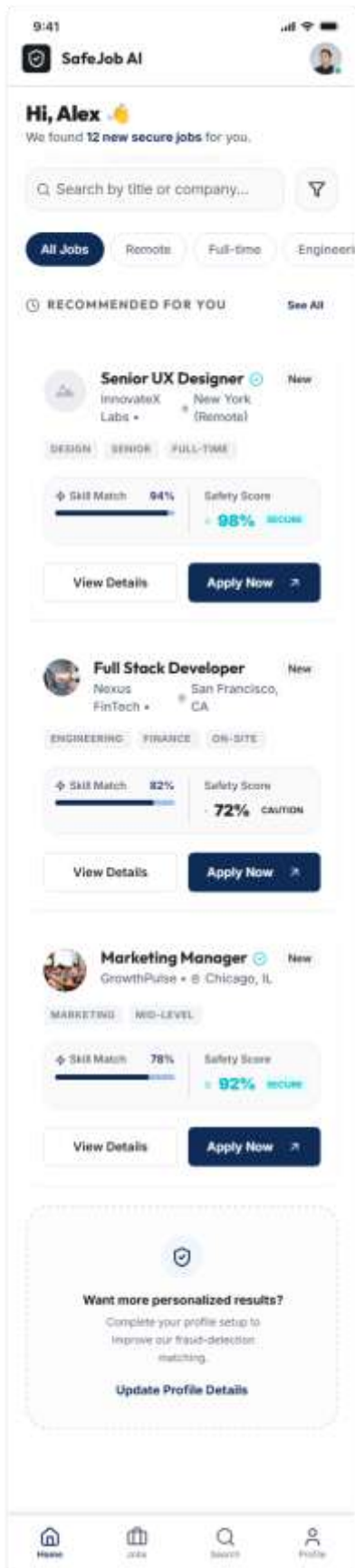
SafeJob Verification:

Your profile data is encrypted and only used to filter out fraudulent job postings that don't match your background.

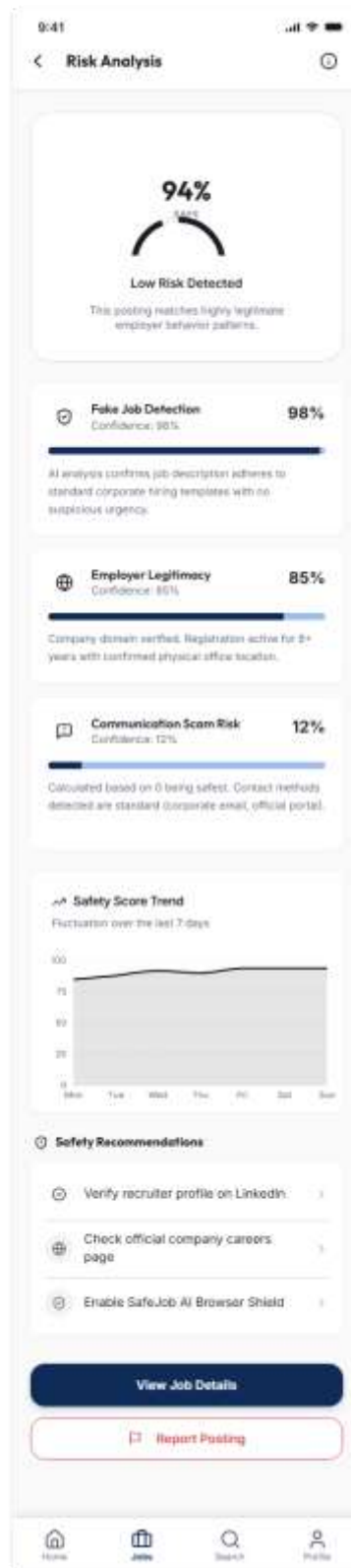
Save Profile & Continue

Skip for now

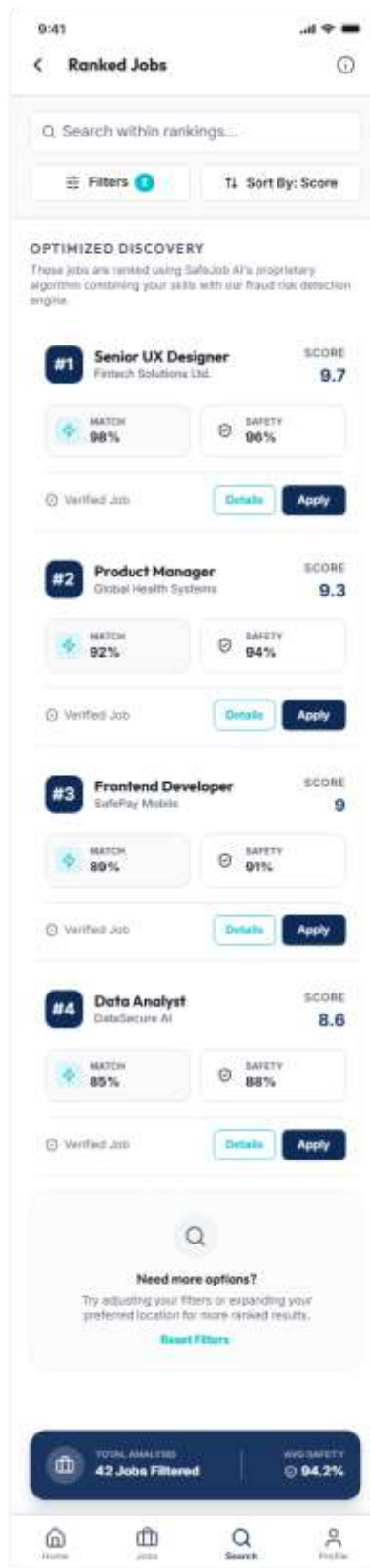
Appendix 5: Profile Setup UI



Appendix 6: Job Recommendation Dashboard



Appendix 7: Job Details Page



Appendix 8: Safe Job Ranking Results Page